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Humanitarian Donations Platform: A Transparent and Accessible System for Managing Verified Cases and Donations

(Graduation Project Part 1) prepared for the Software Engineering course

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Abstract

The Humanitarian Donations Platform is a proposed web-based system designed to help charitable organizations publish documented humanitarian cases and receive donations through a shared digital platform. Each case includes a description, a target amount, and supporting documents intended to help donors understand the case before contributing. The platform is designed for a context in which access to payment infrastructure and digital services may be constrained, particularly in Syria.

A central design decision is that visitors can browse cases and donate without creating an account in advance. After a successful donation, the system either links the donation to an existing donor account or creates an inactive account and sends an activation link. Charities can publish case updates, while donors can receive notifications relevant to the cases they support. The report reviews comparable platforms, identifies a gap in locally suitable multi-charity services, and presents the proposed system's objectives and theoretical foundations, including requirements engineering, use-case modeling, data modeling, UML class diagrams, and data integrity.

Keywords: Humanitarian donations; charitable organizations; verified cases; donor notifications; requirements engineering; ERD; UML; data integrity.

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General Introduction

1.1 Background and Motivation

Digital platforms can make humanitarian giving easier by presenting needs online and allowing donors to contribute remotely. For a donation to be meaningful and trustworthy, however, the donor needs clear information about the individual case, its target amount, and the organization responsible for it. The proposed Humanitarian Donations Platform focuses on documented cases published by charitable organizations, with a shared platform and administrative oversight rather than separate, isolated websites for each organization.

The project is motivated by the needs of the Syrian context described in the project requirements: available payment methods and supporting infrastructure may be limited, and some local charitable work is communicated through social media, spreadsheets, or manual records. A system designed for this setting should be accessible in Arabic, support right-to-left presentation, and reduce unnecessary steps between viewing a case and donating.

1.2 Problem Statement and Research Question

The problem addressed by this project is the lack of a shared, locally appropriate digital platform that combines documented humanitarian cases, multiple charitable organizations, donor follow-up, and reliable donation records. Global crowdfunding services offer useful campaign pages and updates, but the project's review identifies limitations in their suitability for the Syrian context. Some local initiatives provide case-level information, yet they are often operated independently and do not form a common multi-organization service. Manual and semi-digital workflows also make consistent tracking and donor communication more difficult.

Research question: How can a shared humanitarian donation platform be designed to support documented cases from multiple charitable organizations, allow donations without mandatory prior registration, and preserve reliable donation records and donor follow-up in the Syrian context?

1.3 Objectives, Scope and Contribution

The project's objectives are to:

- Provide a common platform on which approved charitable organizations can publish documented humanitarian cases.
- Allow visitors to browse cases and donate without being required to register first.
- After a successful donation, link the donation to an existing account or create an inactive account and send an activation link.
- Support case updates and notifications relevant to donors who contributed to those cases.
- Maintain consistent donation totals and transaction records, including protection against duplicate payment notifications.
- Provide a design that considers Arabic-language presentation and constraints in the local payment and infrastructure environment.

Scope: this project focuses on the preliminary design and theoretical foundations of the humanitarian donations platform. It examines comparable systems, defines the problem and project objectives, outlines the proposed workflow, and explains the software-engineering

concepts guiding the design. At this stage, the work describes the proposed system; it does not claim that a production version or live payment integration has been completed.

Contribution: the proposed design combines a multi-charity structure with case-level documentation, donation-first access, post-donation account linking or creation, donor-focused updates, and explicit data-integrity rules.

1.4 State of the Art

Several existing services provide relevant reference points. The comparison below is based on the systems and criteria documented in the project report. It is not intended to claim that every feature is absent from every platform; rather, it highlights the fit of each model to the requirements of this project.

System	Relevant strengths described in the review	Limitations relative to this project
GoFundMe	Global crowdfunding service with individual campaign pages and creator-published updates.	The reviewed model does not provide the project's approved-charity mediation, full Arabic interface, or practical availability for the Syrian context.
LaunchGood	International crowdfunding focused on Muslim communities, with campaign updates and donor accounts.	Follow-up is centered on the campaign, rather than the project's specific model of documented individual cases and their donor-level updates.
Ehsan (Saudi Arabia)	National charitable platform with digital donations, varied giving opportunities, progress information, notifications and impact reporting.	Its operation is supported by national infrastructure, digital identity and local payment arrangements that cannot simply be assumed in Syria.
Syrian local digital initiatives	Some initiatives publish individual documented cases with targets and collection progress.	The reviewed approach is often organization-specific and does not provide one shared platform with unified administration across multiple charities.
Manual or semi-digital workflows	Paper records, spreadsheets and social-media announcements can be used with limited technical setup.	They do not inherently provide consistent donor accounts, case-specific update tracking, or unified transaction controls.
Proposed platform	Shared multi-charity service, documented cases, donation without prior registration, post-donation account handling, and donor notifications.	These are design goals of the proposed system and require implementation and evaluation before their effectiveness can be established.

The review suggests a gap for a shared platform that is adapted to local constraints while preserving case-level documentation and a consistent administrative model. One distinctive requirement is that the donation itself can be the entry point to an account: after payment succeeds, an existing account is linked or a new inactive account is created and an activation link is sent.

1.5 Proposed Solution

The proposed solution is a web-based platform in which an administrator manages participating charitable organizations and donor accounts, while approved organizations publish documented cases and provide updates. Visitors can view cases, review their

descriptions and supporting documents, and submit donations without first creating an account. The system records the payment result and updates the collected amount only after success has been confirmed.

Following a successful donation, the platform checks whether the donor already has an account. If an account exists, the donation can be linked after the appropriate identity check; otherwise, the system creates an inactive account and sends an activation link. The design also includes case-specific notifications, administrative controls, logical deletion where transaction history must be retained, and safeguards against counting the same payment more than once. Payment-provider integration and local deployment details remain implementation concerns to be validated against available services.

Chapter 1: Theoretical Study

1.6 Humanitarian Donation Platforms and Transparency

A humanitarian donation platform connects donors with needs presented by charitable organizations. In this project, transparency is supported by displaying an individual case description, a target amount and supporting documents before donation. The platform also separates the organization responsible for publishing a case from the administrator responsible for managing organizations on the shared service. These design choices aim to make the origin and status of a case easier to understand; they do not replace the need for a real verification and governance process.

Case updates provide a way for organizations to communicate progress after publication. Donor-focused notifications can make those updates easier to follow, particularly when they relate to a case supported by a donor. The reviewed examples—GoFundMe, LaunchGood and Ehsan—illustrate campaign pages, updates, donor accounts or reporting features, while the proposed platform adapts selected ideas to its stated local context and multi-charity model.

1.7 Requirements Engineering

Requirements engineering defines what a system must do and the quality constraints it must satisfy. Functional requirements describe system services, while non-functional requirements describe properties such as security, reliability, usability and performance. The project's requirements distinguish user-facing activities—browsing cases, donating, account activation and receiving updates—from organization and administration activities such as publishing cases and managing participating charities.

Requirements should be clear, testable and traceable to design decisions. For example, the requirement to permit donation without prior registration leads to a donation flow that accepts contact details before account creation. The requirement to preserve accurate totals leads to a rule that the collected amount is changed only when a payment transitions to a confirmed successful state. The project report identifies that quantitative performance targets still need to be specified before they can be used as measurable acceptance criteria.

1.8 Use-Case Modeling

A use-case model describes how external actors interact with a system to achieve goals. The project identifies visitors and donors, charitable organizations, an administrator, a payment provider and a notification service as relevant actors or external participants. Use cases include browsing cases, donating, confirming payment, publishing a case, publishing updates, sending notifications, activating an account and administering organizations or donor accounts.

The donation flow is central to the model: the visitor selects a case, provides the donation amount and required contact information, selects a payment method, and the system processes and verifies the payment. After success, the transaction is recorded and the case total is updated. Account handling then follows one of two paths: link the donation to an existing account or create an inactive account and send an activation link. These paths are alternatives, not simultaneous actions.

Use-case relationships should reflect required behavior rather than merely visual convenience. An «include» relationship represents behavior required by the base use case, whereas

«extend» represents conditional or optional behavior. The model also treats login as a precondition for protected organization and administrator actions. This makes access assumptions explicit and helps connect user-visible behavior to requirements.

1.9 Relational Data Modeling and ERD

An Entity–Relationship Diagram (ERD) represents the principal data entities and the relationships between them before or alongside relational database implementation. The project's data model contains ten entities, including users, roles, charitable organizations, humanitarian cases, donations, payment records and notifications. The model is intended to support the use cases while keeping donation history consistent when accounts or organizations are later deactivated or logically deleted.

A key design decision is to allow a donation to exist before it is associated with a registered donor account. The record can retain donor contact information, and after successful payment it can be linked to an existing account or a newly created one. The design also separates the donation from the payment-provider reference so that transaction processing can be checked independently. Proposed state values for organizations, cases, donations and payment-related records should be constrained in the database, for example through CHECK constraints or enumerated types where supported.

1.10 UML Class Diagrams

A UML class diagram describes the static structure of a system through classes, attributes, operations and relationships. In the proposed platform, the class model is intended to align with the data model while also representing responsibilities that may exist in application logic. Examples include managing users and roles, publishing cases, recording donations, processing payment outcomes, and preparing notifications.

The class relationships and multiplicities should be consistent with the underlying business rules. For example, a charitable organization can publish multiple cases, a case can receive multiple donations, and a donor can have a history of donations. The design also distinguishes a user's role from the organization and case records that the user is permitted to manage. Keeping the class diagram aligned with the ERD reduces the risk of contradictory assumptions between the application and database layers.

1.11 Data Integrity, Privacy and Payment Reliability

Donation systems must protect both financial consistency and personal information. The project specifies that duplicate payment-provider notifications must not cause a donation to be counted more than once. A unique constraint on the provider and provider reference can prevent duplicate payment records, but it is not sufficient by itself: the application must also allow the payment state to transition to "succeeded" only once and update the case's collected amount in the same database transaction.

Before a donation is accepted as successful, the confirmed amount and currency should match the recorded donation, and the donation currency should match the case currency. Notifications that arrive while a payment is still pending must not increase the collected amount until success is confirmed. These rules make payment processing idempotent, meaning that repeating the same notification does not repeat its financial effect.

The project also proposes logical deletion to preserve the integrity of donation history. When a donor account is deleted, identifying fields can be anonymized while the transaction record

remains available for correct accounting. Similarly, an organization with active cases should not be physically removed without an appropriate workflow. Access to administrative and organization-management functions should require authentication and authorization. These are design requirements; their effectiveness depends on careful implementation and testing.

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